

3MTT CYBERSECURITY PROGRAM

CAPSTONE PROJECT REPORT

Topic : Wi-Fi Security Assessment (Security Assessment of an MTN ZLT M305 Pro 4G MiFi)

Author: Ismaeel Ekundayo

Program: 3MTT Cybersecurity

Assessment Environment: virtual environment (virtualbox &kali Linux)

Target Device: MTN ZLT M305 Pro MiFi

Target IP Address: 192.168.0.1

Assessment Host: Windows Host / Kali Linux VM

EXECUTIVE SUMMARY

This project presents a security assessment of an MTN ZLT M305 Pro 4G MiFi conducted as part of the 3MTT Cybersecurity Program Capstone Project.

The objective was to assess the security posture of the wireless network and its administrative interface using a controlled laboratory environment. The assessment combined network reconnaissance, service enumeration, packet analysis, wireless configuration review, and web-interface security assessment.

The primary tools used were Nmap, Wireshark, and OWASP ZAP.

The wireless configuration was found to use WPA2-PSK with AES encryption, and the default administrator password had been changed. No configurable WPS functionality was identified during the review. The device firmware version observed during the assessment was 3.06.05.

Network traffic was analyzed using Wireshark on the Windows host to observe communication between the assessment devices and the MiFi.

OWASP ZAP was used to passively assess the MiFi's HTTP-based management interface. Several security observations were identified, including the absence of a Content Security Policy header, missing security-related HTTP headers, an outdated JavaScript library identified by ZAP, and several information-disclosure observations.

Nmap was also used to assess the Windows assessment host. Services including MSRPC, NetBIOS/SMB, and HTTPAPI were identified. The vulnerability scripts did not confirm the tested SMB vulnerabilities.

Overall, the assessment did not demonstrate a successful compromise of the MiFi. However, the findings indicate areas where the device's management interface and connected client environment could be further hardened.

INTRODUCTION

Wireless networks are widely used in homes, offices, schools, and small businesses. While wireless connectivity improves accessibility and mobility, poorly configured Wi-Fi equipment can expose users to unauthorized access, unnecessary services, weak authentication, and vulnerable management interfaces.

This project focuses on assessing the security posture of an MTN ZLT M305 Pro MiFi in a controlled and authorized laboratory environment.

The assessment demonstrates a practical cybersecurity workflow:

1. Identify devices and services.
2. Review wireless security configuration.
3. Observe network traffic.
4. Assess the router's management interface.
5. Document security findings.
6. Recommend appropriate remediation and hardening measures.

PROJECT OBJECTIVES

The objectives of the assessment were to:

- Identify active devices within the authorized lab network.
- Identify exposed network services.
- Review the MiFi's wireless security configuration.
- Observe network traffic generated within the lab.
- Assess the MiFi's web-based administrative interface.
- Identify security weaknesses and potential information disclosures.
- Evaluate the security posture of an associated Windows client.
- Provide practical remediation and hardening recommendations.
- Produce evidence suitable for a cybersecurity assessment report.

SCOPE AND AUTHORIZATION

Scope

The assessment was limited to the authorized laboratory environment consisting of:

- MTN ZLT M305 Pro MiFi
- Kali Linux virtual machine
- Windows host
- Authorized devices connected to the MiFi

The primary assessment target was the MiFi and its local management interface.

The Windows host was also assessed as a connected client to determine whether unnecessary services were exposed to other devices on the local network.

Authorization

The assessment was performed on personally controlled equipment within an authorized laboratory environment.

No third-party networks, systems, or devices were intentionally targeted.

Testing was conducted with the objective of learning defensive security assessment techniques and identifying opportunities for hardening.

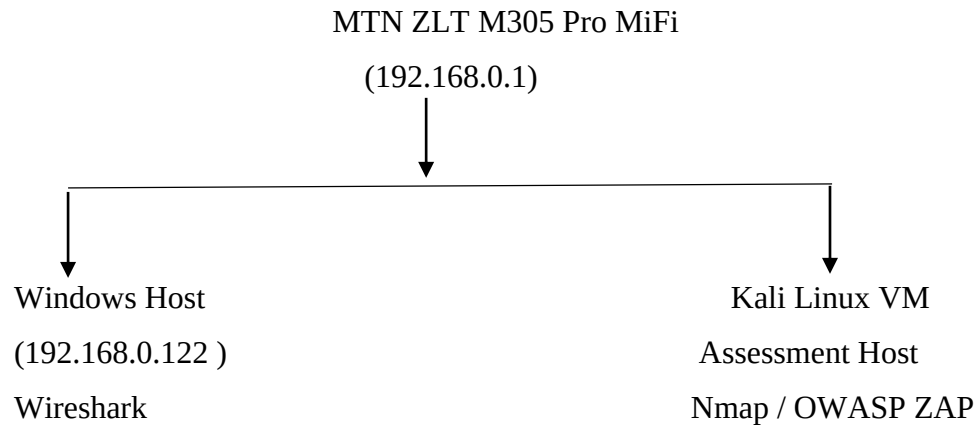
LAB ENVIRONMENT

The laboratory consisted of an MTN ZLT M305 Pro MiFi providing the local wireless network.

The Kali Linux virtual machine was connected to the same network and used for Nmap and OWASP ZAP testing.

Wireshark was operated on the Windows host for packet capture and network traffic analysis.

Logical Lab Architecture



The laboratory was isolated from unauthorized external systems as far as practical.

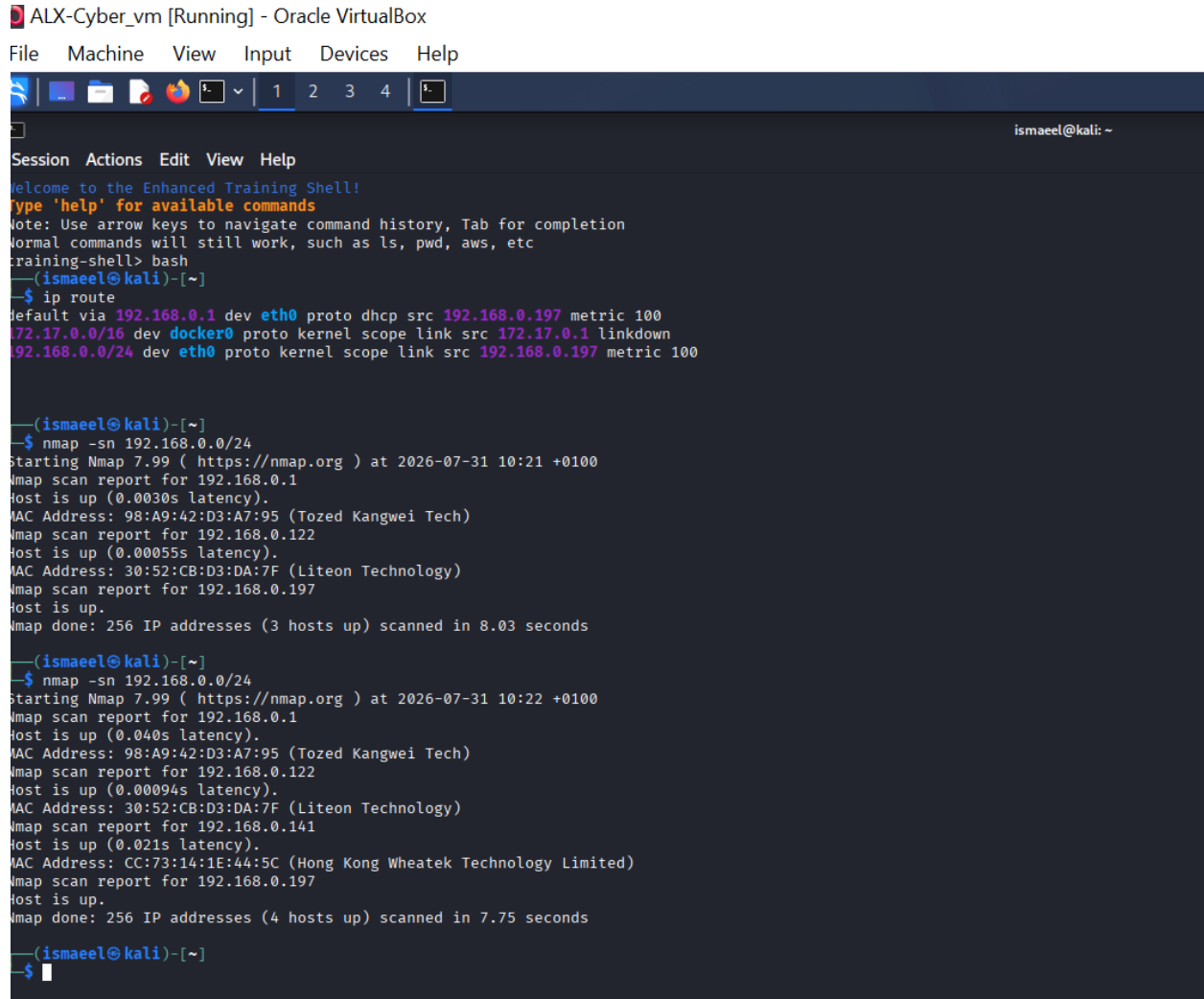
TOOLS USED

Tool	Purpose
Nmap 7.99	Network discovery, service enumeration and NSE vulnerability checks
Wireshark 4.4.7	Packet capture and network traffic analysis
OWASP ZAP 2.17.0	Passive assessment of the MiFi web management interface
Kali Linux	Security assessment workstation
VirtualBox	Virtual laboratory environment

METHODOLOGY

The assessment followed a phased approach.

Phase 1 : Network Discovery



ALX-Cyber_vm [Running] - Oracle VirtualBox

File Machine View Input Devices Help

ismaeel@kali: ~

```
Session Actions Edit View Help
Welcome to the Enhanced Training Shell!
Type 'help' for available commands
Note: Use arrow keys to navigate command history, Tab for completion
Normal commands will still work, such as ls, pwd, aws, etc
training-shell> bash
--(ismaeel@kali)-[~]
$ ip route
default via 192.168.0.1 dev eth0 proto dhcp src 192.168.0.197 metric 100
172.17.0.0/16 dev docker0 proto kernel scope link src 172.17.0.1 linkdown
192.168.0.0/24 dev eth0 proto kernel scope link src 192.168.0.197 metric 100

--(ismaeel@kali)-[~]
$ nmap -sn 192.168.0.0/24
Starting Nmap 7.99 ( https://nmap.org ) at 2026-07-31 10:21 +0100
Nmap scan report for 192.168.0.1
Host is up (0.0030s latency).
MAC Address: 98:A9:42:D3:A7:95 (Tozed Kangwei Tech)
Nmap scan report for 192.168.0.122
Host is up (0.00055s latency).
MAC Address: 30:52:CB:D3:DA:7F (Liteon Technology)
Nmap scan report for 192.168.0.197
Host is up.
Nmap done: 256 IP addresses (3 hosts up) scanned in 8.03 seconds

--(ismaeel@kali)-[~]
$ nmap -sn 192.168.0.0/24
Starting Nmap 7.99 ( https://nmap.org ) at 2026-07-31 10:22 +0100
Nmap scan report for 192.168.0.1
Host is up (0.040s latency).
MAC Address: 98:A9:42:D3:A7:95 (Tozed Kangwei Tech)
Nmap scan report for 192.168.0.122
Host is up (0.00094s latency).
MAC Address: 30:52:CB:D3:DA:7F (Liteon Technology)
Nmap scan report for 192.168.0.141
Host is up (0.021s latency).
MAC Address: CC:73:14:1E:44:5C (Hong Kong Wheatek Technology Limited)
Nmap scan report for 192.168.0.197
Host is up.
Nmap done: 256 IP addresses (4 hosts up) scanned in 7.75 seconds

--(ismaeel@kali)-[~]
$
```

Figure 1: Network and Devices Discovery

Nmap was used to identify active hosts on the local network. The assessment focused only on the authorized lab subnet.

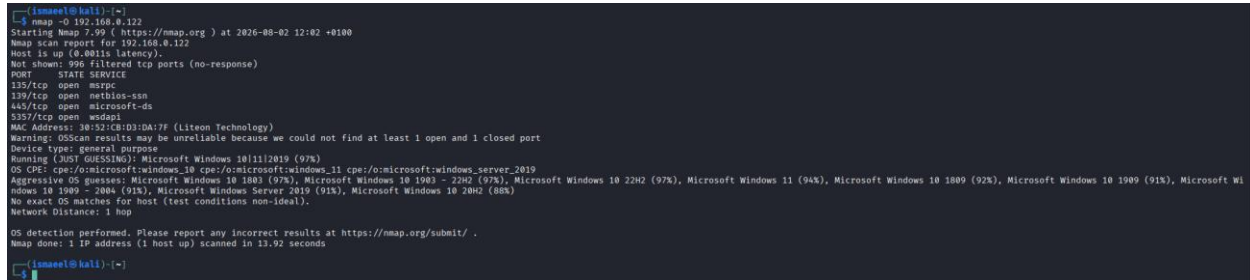
Host discovery was performed using Nmap's ping-scan functionality.

Ip route

nmap -sn 192.168.0.0/24

The scan identified the MiFi gateway and connected assessment devices.

Phase 2 — Service Enumeration



```
(ismael@kali)~$ nmap -O 192.168.0.122
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-02 12:02 -0100
Nmap scan report for 192.168.0.122
Host is up (0.0011s latency).
Not shown: 996 filtered tcp ports (no-response)
PORT      STATE SERVICE
135/tcp    open  msrpc
139/tcp    open  netbios-ssn
445/tcp    open  microsoft-ds
5357/tcp   open  wsdapi
MAC Address: 38:52:C8:D3:0A:7F (Liteon Technology)
Warning: OSscan results may be unreliable because we could not find at least 1 open and 1 closed port
Device type: general purpose
Running (JUST GUESSING): Microsoft Windows 10|11|2019 (97%)
OS CPE: cpe:/o:microsoft:windows.10 cpe:/o:microsoft:windows.11 cpe:/o:microsoft:windows_server.2019
Aggressive OS guesses: Microsoft Windows 10 1803 (97%), Microsoft Windows 10 1803 - 22H2 (97%), Microsoft Windows 11 (94%), Microsoft Windows 10 1809 (92%), Microsoft Windows 10 1909 (91%), Microsoft Wi
ndows 10 1909 - 20H4 (91%), Microsoft Windows Server 2019 (91%), Microsoft Windows 10 20H2 (88%)
No exact OS matches for host (test conditions non-ideal).
Network Distance: 1 hop

OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 13.92 seconds

(ismael@kali)~$
```

Figure 2: Service Enumeration for Windows Host

Nmap service detection was used to identify services exposed by the Windows assessment host.

The assessment identified services associated with:

- TCP/135 — MSRPC
- TCP/139 — NetBIOS Session Service
- TCP/445 — Microsoft-DS/SMB
- TCP/5357 — HTTPAPI

These services were reviewed to determine whether they represented unnecessary exposure within the local network.

Phase 3 : Vulnerability Assessment of Windows Client

```
(ismaeel@kali)-[~]
$ nmap -sV --script vuln 192.168.0.122
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-02 12:27 +0100
Nmap scan report for 192.168.0.122
Host is up (0.00085s latency).
Not shown: 996 filtered tcp ports (no-response)
PORT      STATE SERVICE      VERSION
135/tcp    open  msrpc        Microsoft Windows RPC
139/tcp    open  netbios-ssn  Microsoft Windows netbios-ssn
445/tcp    open  microsoft-ds?
5357/tcp   open  http         Microsoft HTTPAPI httpd 2.0 (SSDP/UPnP)
|_http-server-header: Microsoft-HTTPAPI/2.0
|_http-csrf: Couldn't find any CSRF vulnerabilities.
|_http-stored-xss: Couldn't find any stored XSS vulnerabilities.
|_http-dombased-xss: Couldn't find any DOM based XSS.
MAC Address: 30:52:CB:D3:DA:7F (Liteon Technology)
Service Info: OS: Windows; CPE: cpe:/o:microsoft:windows

Host script results:
|_smb-vuln-ms10-054: false
|_smb-vuln-ms10-061: Could not negotiate a connection:SMB: Failed to receive bytes: ERROR
|_samba-vuln-cve-2012-1182: Could not negotiate a connection:SMB: Failed to receive bytes: ERROR

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 185.68 seconds

(ismaeel@kali)-[~]
$
```

Figure 3: vulnerability assessment of windows client

A safe Nmap NSE vulnerability scan was performed against the authorized Windows host:

```
nmap -sV --script vuln 192.168.0.122
```

The scan did not confirm the tested SMB vulnerabilities.

In particular, the results included:

- smb-vuln-ms10-054: false
- Other SMB vulnerability checks returned negotiation errors rather than a positive vulnerability result.

The negotiation errors were not interpreted as proof of vulnerability.

The exposed SMB services were instead treated as an attack-surface observation requiring appropriate firewall and service-management controls.

WIRELESS SECURITY CONFIGURATION REVIEW

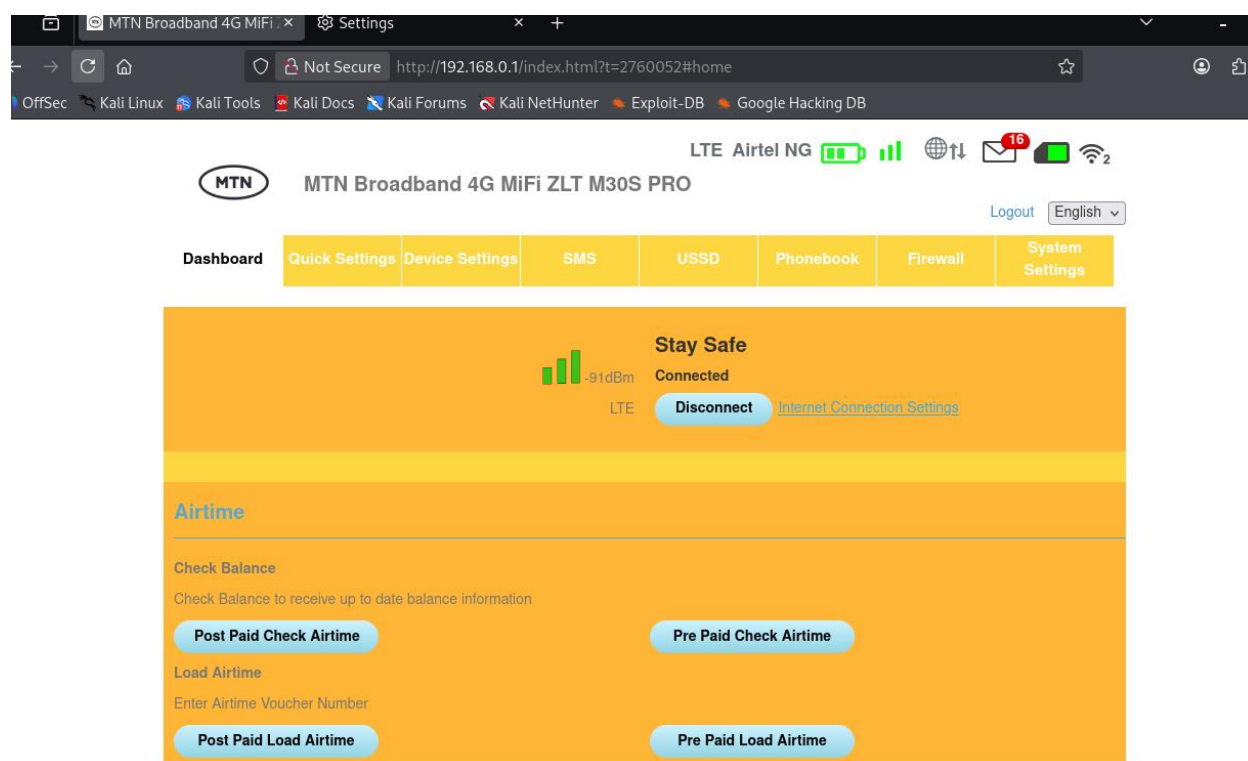


Figure 4: Mifi device interface

The MiFi administration interface was reviewed manually.

The following configuration information was recorded.

Security Item	Observation	Assessment
Wireless security	WPA2-PSK	Positive
Encryption	AES	Positive
Administrator password	Default password changed	Positive
Firmware	Version 3.06.05	Monitor for updates
Connected devices	2 during assessment	Expected in lab

Security Item	Observation	Assessment
WPS	No configurable WPS feature identified	Positive/Reduced exposure

WPA2-PSK/AES

The MiFi was configured with WPA2-PSK using AES encryption.

This provides a significantly stronger wireless security baseline than obsolete configurations such as WEP or WPA using legacy encryption.

Where supported by the device, WPA3 should be considered as a future improvement.

Administrator Password

The default administrator password had been changed.

This is an important security control because retaining default credentials can allow unauthorized users with network access to gain administrative control.

WPS

No WPS option or WPS functionality was identified through the device's management interface or physical controls during the assessment.

This reduces exposure to risks associated with WPS-based authentication mechanisms.

The finding is recorded as:

No configurable WPS functionality identified during assessment

NETWORK TRAFFIC ANALYSIS

Wireshark 4.4.7 was used on the Windows host to observe traffic generated within the authorized assessment environment.

The capture was used to examine common protocols and communication patterns, including:

- ARP
- DNS
- ICMP
- TCP
- TLS/HTTPS

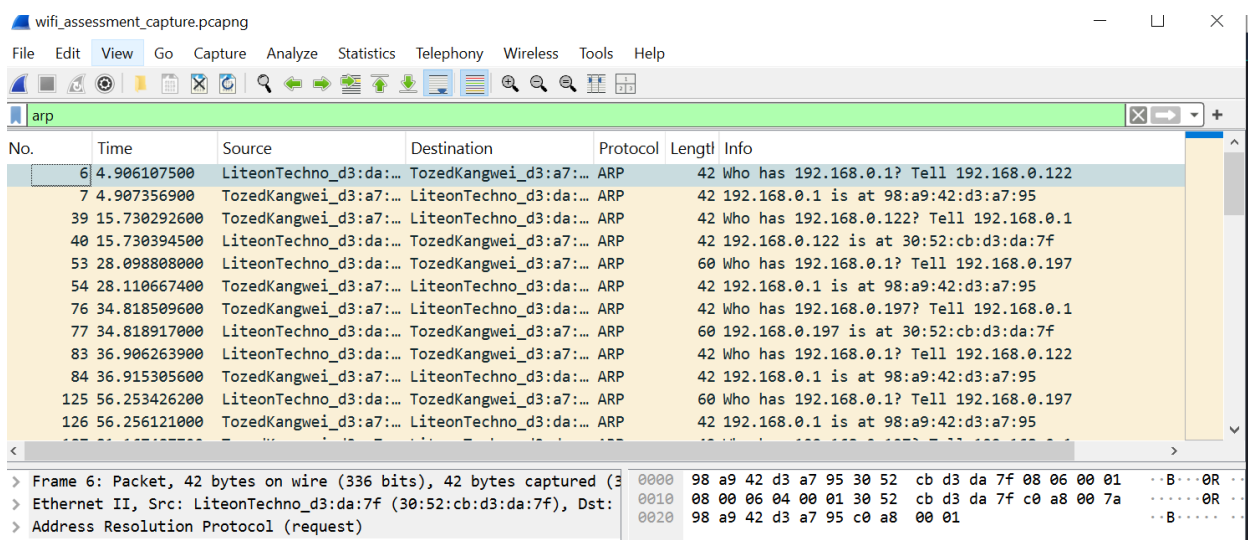


Figure 5: Wireshark packet filter using ARP

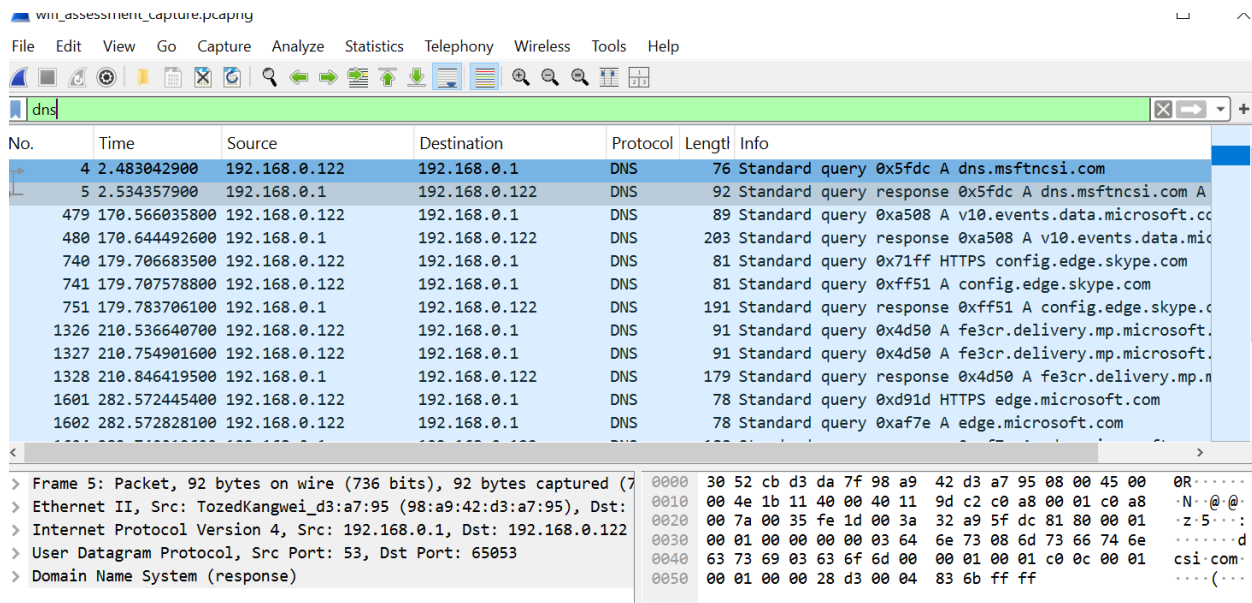


Figure 6:Wireshark packet filter using DNS

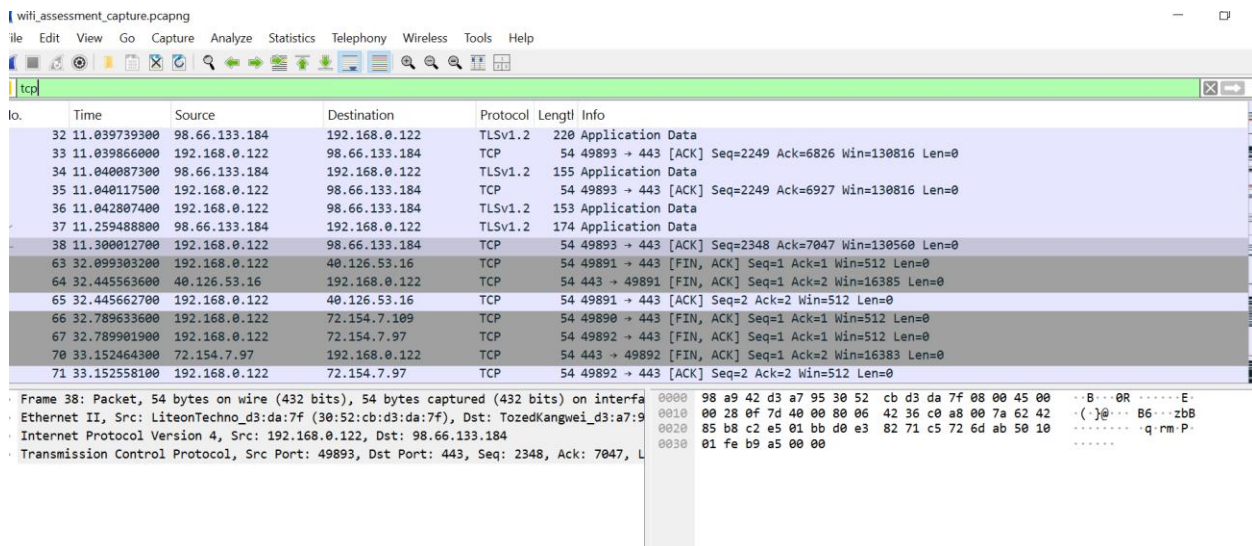


Figure 7:Wireshark packet filter using TCP

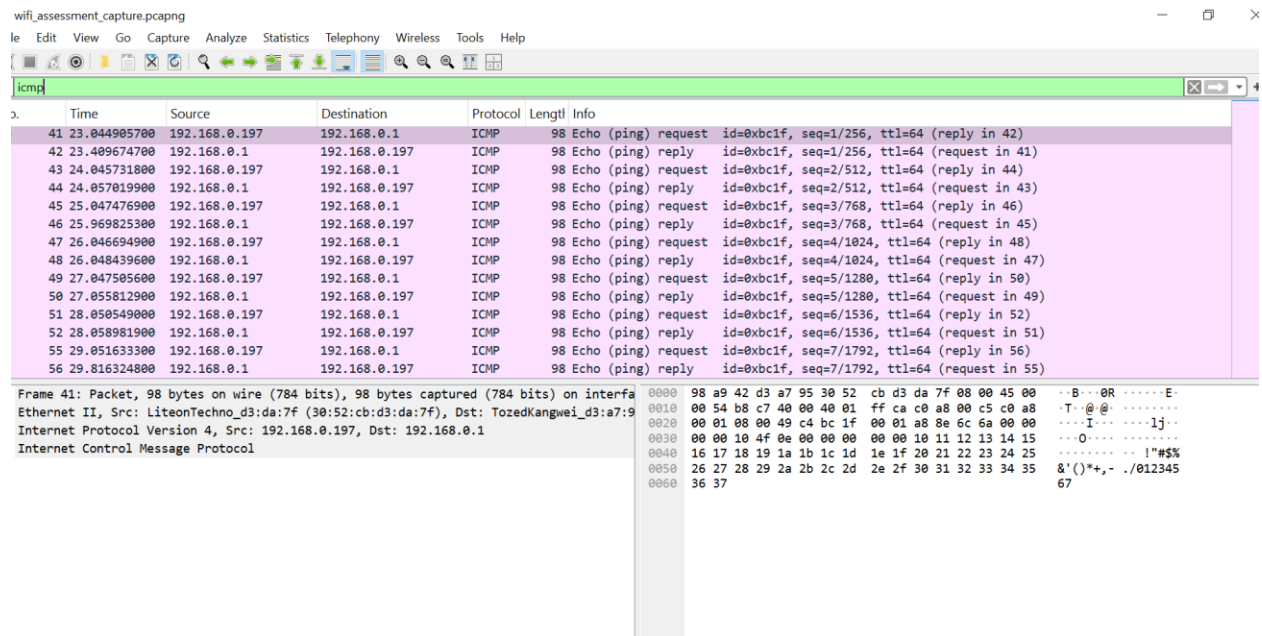


Figure 8: Wireshark packet filter using ICMP

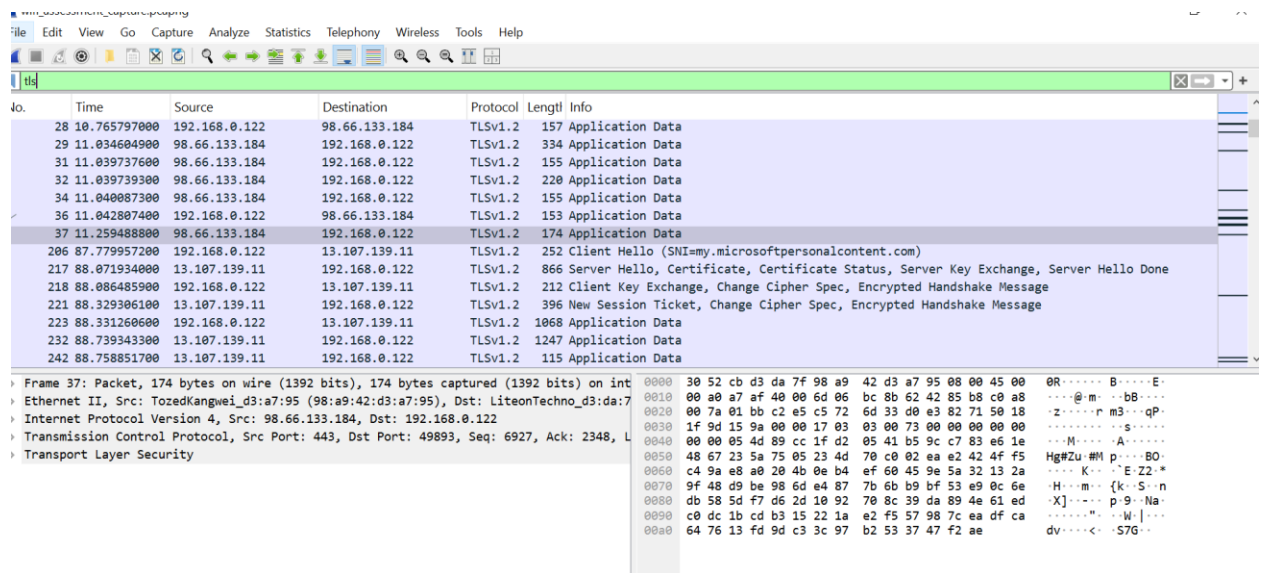


Figure 9: Wireshark packet filter using TLS/HTTP

The analysis demonstrated normal local-network communication between the assessment devices and the gateway.

The Wireshark component of the assessment was primarily observational and was not used to intercept or decrypt protected communications.

10. OWASP ZAP Web Interface Assessment

OWASP ZAP version 2.17.0 was installed on Kali Linux.

The MiFi's web management interface was accessed through the ZAP proxy:

http://192.168.0.1

The assessment initially focused on passive scanning.

The MiFi management pages were browsed through a browser configured to use ZAP as a proxy. ZAP successfully captured the HTTP requests and responses.

The passive assessment produced several alerts.

ZAP FINDINGS

Finding ZAP-01 — Content Security Policy Header Not Set

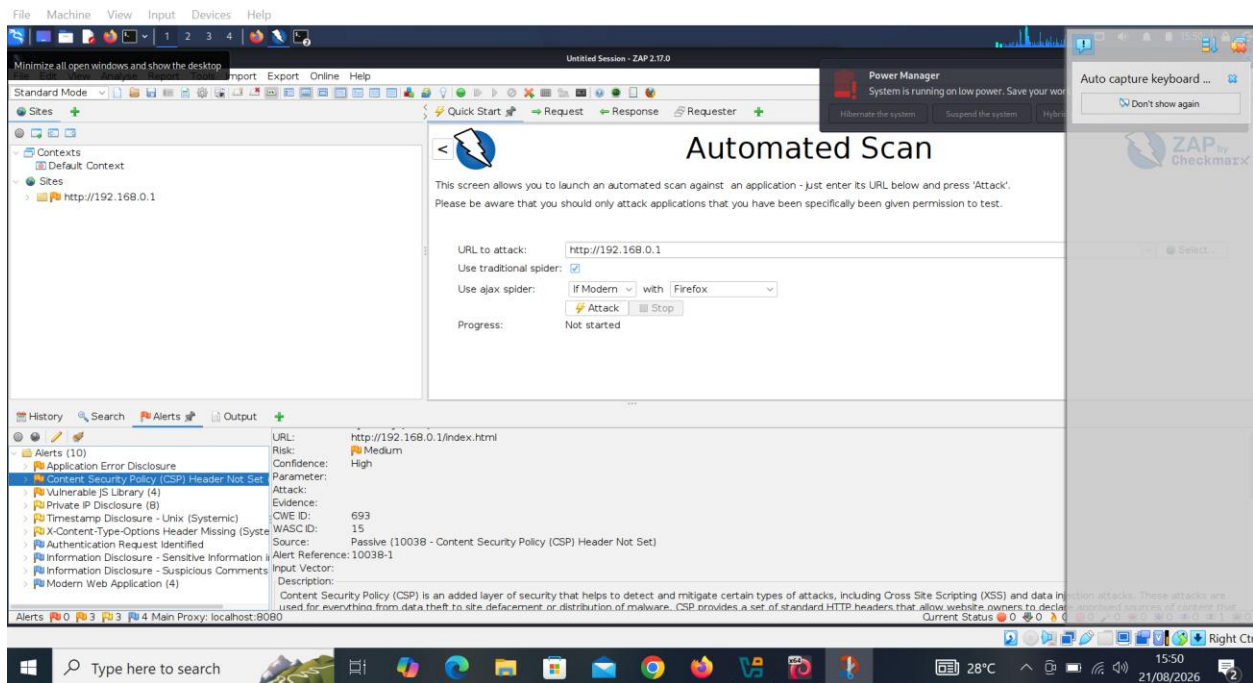


Figure 10: OWASP ZAP interface

Severity reported by ZAP: Medium

Description

The MiFi's web management interface did not provide a Content Security Policy (CSP) HTTP response header.

CSP is a browser security control that can reduce the impact of certain classes of web attacks, particularly when combined with other vulnerabilities such as cross-site scripting.

Impact

The absence of CSP does not by itself demonstrate that the MiFi is vulnerable to XSS or that the device can be compromised.

However, it represents a missing defense-in-depth control.

Recommendation

If supported by the device's firmware and web server, the manufacturer should implement an appropriate Content Security Policy.

For the device owner, the practical mitigation is to:

- Keep firmware updated.
- Restrict access to the management interface.
- Avoid exposing the management interface to untrusted networks.

Risk: Medium

Finding ZAP-02 : Vulnerable JavaScript Library

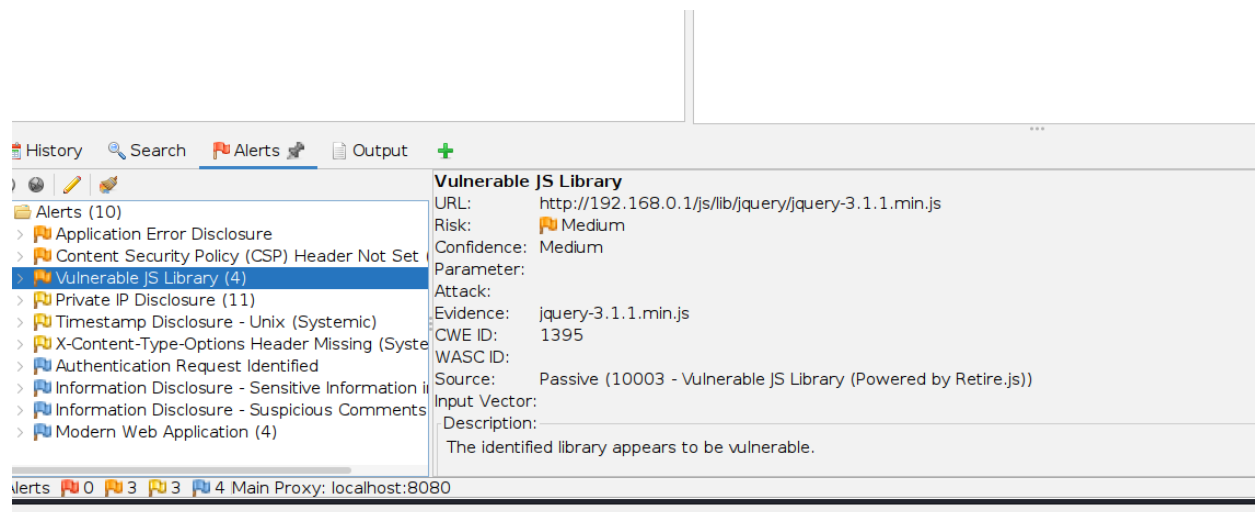


Figure 11: Vulnerable JavaScript Library i

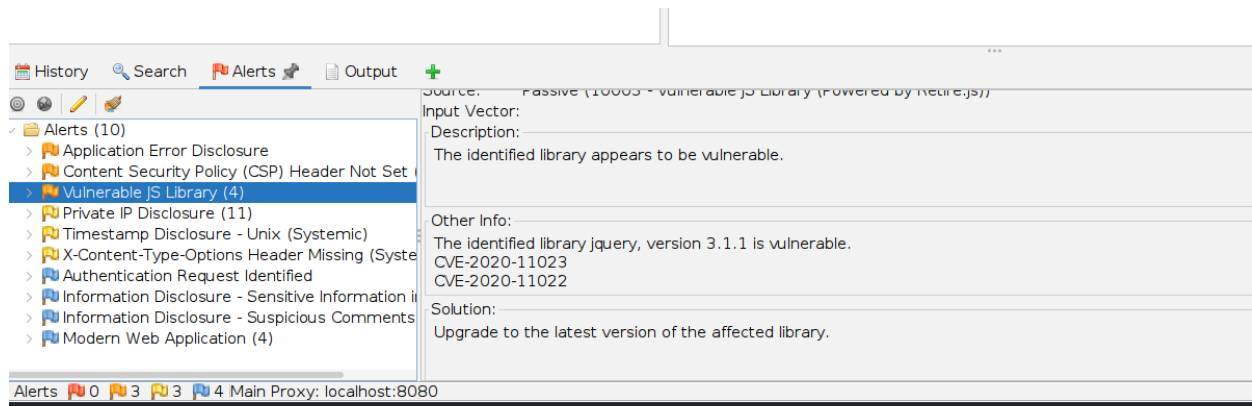


Figure 12: Vulnerable JavaScript Library ii

Severity reported by ZAP: Medium

Description

ZAP identified the use of **jQuery version 3.1.1** in the MiFi web interface.

ZAP associated this library version with known vulnerabilities including:

- CVE-2020-11022
- CVE-2020-11023

Impact

The presence of an outdated JavaScript library increases the software attack surface.

However, identifying an outdated library does not by itself prove that the MiFi is exploitable.

Exploitability depends on how the library is used by the application and whether an attacker can reach a vulnerable code path.

Recommendation

The device manufacturer should update the bundled JavaScript library to a supported version and test the management interface for compatibility.

For the device owner:

- Maintain current firmware.
- Monitor vendor firmware releases.
- Restrict administrative interface access to trusted users.

Risk: Medium

Finding ZAP-03 : X-Content-Type-Options Header Missing

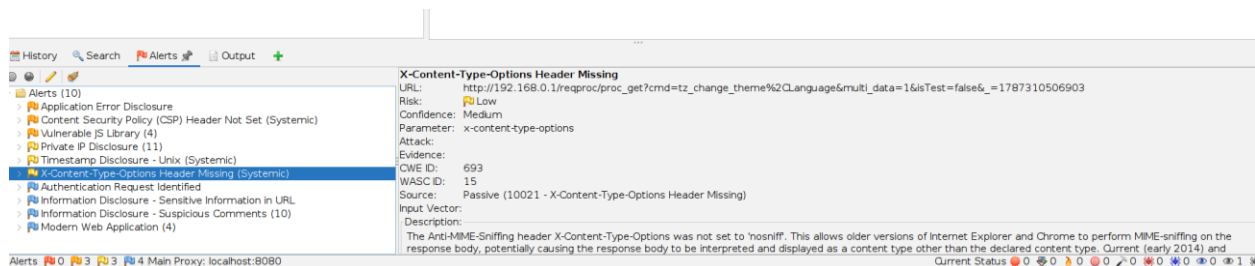


Figure 13:X-Content-Type-Options Header Missing

Description

ZAP identified that the HTTP response did not include the recommended X-Content-Type-Options security header.

Impact

This is a defense-in-depth weakness that can allow browsers to perform MIME-type sniffing under certain circumstances.

Recommendation

The device manufacturer should configure the web server to return:

X-Content-Type-Options: nosniff

Risk: Low/Medium

Finding ZAP-04 : Information Disclosure

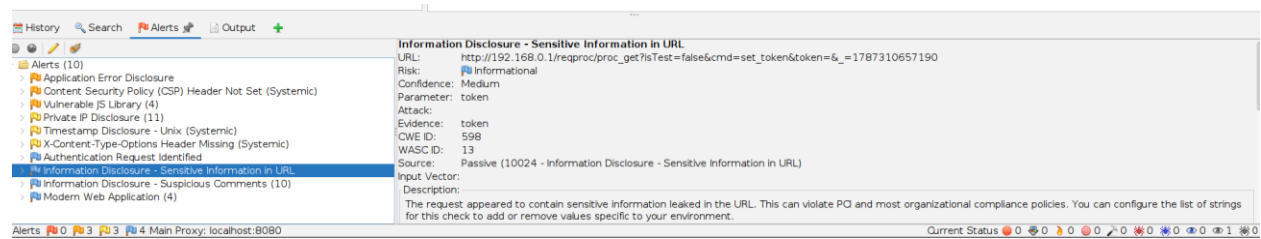


Figure 14:Information Disclosure

ZAP identified information disclosure observations in the management interface.

Information disclosure can provide useful contextual information to an attacker attempting to understand the underlying application.

Recommendation

The manufacturer should minimize unnecessary technical information returned by the management interface.

Risk: Low

Finding ZAP-05 : Timestamp Disclosure

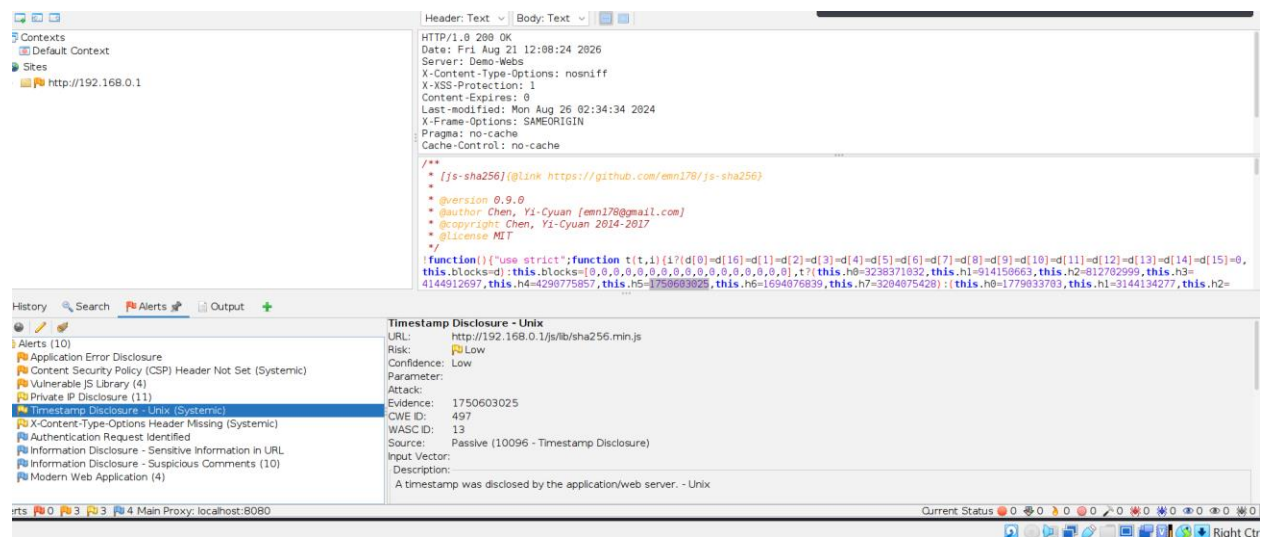


Figure 15:Timestamp Disclosure

ZAP identified timestamp information in the web interface.

Impact

Timestamp information generally represents a low-level information disclosure issue. In isolation, it is unlikely to result in compromise.

Recommendation

Remove unnecessary timestamps from responses where they do not serve a legitimate operational purpose.

Risk: Low

Finding ZAP-06 : Application Error Disclosure

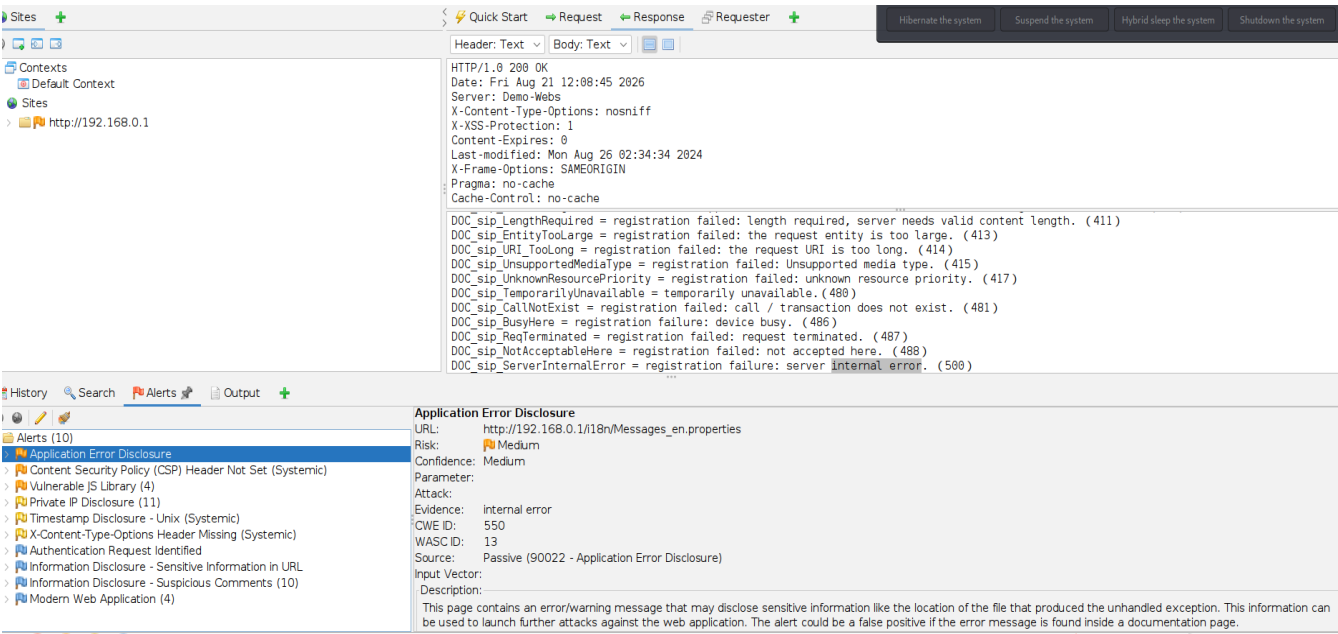


Figure 16:Application Error Disclosure

ZAP identified application error-related information.

Impact

Verbose error messages can reveal implementation details that may assist an attacker during reconnaissance.

Recommendation

The web interface should return generic user-facing error messages while logging detailed diagnostic information internally.

Risk: Low

WINDOWS CLIENT EXPOSURE

The Windows assessment host exposed several services:

Port	Service	Security Consideration
135	MSRPC	Restrict to trusted local network devices
139	NetBIOS/SMB	Disable if unnecessary
445	SMB	Restrict with Windows Firewall
5357	HTTPAPI	Review whether required

The presence of these services does not automatically indicate compromise.

However, unnecessary network services increase the attack surface of a client device.

Recommendation

If file and printer sharing is not required:

- Disable unnecessary sharing services.

- Restrict SMB through Windows Defender Firewall.
- Keep Windows fully patched.
- Avoid exposing SMB services to untrusted networks.

13. Risk Summary

ID	FINDING	SEVERITY	STATUS
ZAP-01	CSP header missing	Medium	Remediation recommended
ZAP-02	Outdated jQuery 3.1.1	Medium	Firmware/vendor remediation recommended
ZAP-03	X-Content-Type-Options missing	Low/Medium	Hardening recommended
ZAP-04	Information disclosure	Low	Hardening recommended
ZAP-05	Timestamp disclosure	Low	Hardening recommended
ZAP-06	Application error disclosure	Low	Hardening recommended
NET-01	SMB exposed on Windows client	Medium	Restrict/disable if unnecessary
NET-02	SMRPC exposed on Windows client	Low/Medium	Restrict to trusted network

No critical vulnerability or successful compromise was demonstrated during the assessment.

REMEDIATION AND HARDENING RECOMMENDATIONS

MiFi Hardening

1. Maintain strong Wi-Fi authentication

Continue using WPA2-PSK with AES.

If the device supports WPA3, consider upgrading to WPA3.

2. Use a strong Wi-Fi password

The wireless password should be unique and resistant to guessing.

A long passphrase is recommended.

3. Maintain a unique administrator password

The default administrator password has already been changed.

Continue using a strong, unique administrative password.

4. Keep firmware updated

Regularly check for firmware updates from the device vendor or network provider.

Firmware updates may address security vulnerabilities and outdated software components.

5. Restrict administrative access

The MiFi management interface should only be accessible from trusted devices.

Remote management should be disabled if it is not required.

6. Review connected devices

Periodically review devices connected to the MiFi and investigate unfamiliar devices.

7. Address web-interface weaknesses

Where supported by future firmware:

- Implement CSP.
- Add appropriate security headers.
- Update outdated JavaScript libraries.
- Reduce information disclosure.
- Avoid verbose error messages.

WINDOWS CLIENT HARDENING

The Windows host should also be hardened because it is connected to the same wireless network.

Recommended measures include:

- Disable SMB/file sharing if not required.
- Restrict TCP/445 using Windows Firewall.
- Restrict NetBIOS where possible.
- Review RPC exposure.
- Keep Windows security updates current.
- Enable and maintain the host firewall.
- Avoid connecting to untrusted public Wi-Fi while file sharing is enabled.

LIMITATIONS

The assessment had several limitations.

1. The assessment was conducted against a consumer MiFi device rather than an enterprise wireless controller.
2. The MiFi's firmware may limit access to advanced security configuration options.
3. No source code was available for the MiFi web application.
4. OWASP ZAP passive findings were not treated as proof of exploitability.
5. The Nmap vulnerability scan was conducted against the authorized Windows client rather than the MiFi itself.
6. No destructive exploitation or password-cracking activity was performed.

7. Wireshark was used for observational traffic analysis; encrypted traffic was not decrypted.
8. The assessment represents a point-in-time security review. Firmware or configuration changes may alter the results.

CONCLUSION

The security assessment demonstrated a structured approach to evaluating the security posture of an MTN ZLT M305 Pro MiFi and an associated Windows client within an authorized laboratory environment.

The wireless configuration demonstrated several positive security characteristics, including WPA2-PSK with AES encryption, a changed administrator password, and no identified configurable WPS functionality.

Network enumeration identified several services exposed by the Windows assessment host, particularly SMB and MSRPC. These services did not demonstrate a confirmed critical vulnerability during the Nmap NSE assessment, but they represent an attack surface that should be restricted when not required.

The most significant observations came from the MiFi's web management interface. OWASP ZAP identified missing security headers, an outdated jQuery library, and several information-disclosure-related observations.

These findings do not demonstrate that the MiFi was successfully compromised. Instead, they highlight opportunities for defense-in-depth improvements and firmware-level remediation.

Overall, the MiFi demonstrated a reasonable baseline wireless security configuration, while its web management interface and connected client environment would benefit from additional hardening.

ETHICAL STATEMENT

All security testing described in this report was conducted within an authorized laboratory environment.

The assessment was performed for educational and defensive cybersecurity purposes as part of the 3MTT Cybersecurity Program Capstone Project.

No unauthorized third-party systems were intentionally targeted, and no destructive testing was performed.